

Boonyakorn 'Than' Thanpanit

I am an EECS student at MIT interested in embedded systems and FPGAs. In addition to formal coursework, I have more than a decade of experience as an electronics hobbyist. In high school, I won a gold medal at IOAA, a silver medal at IPhO, and a bronze medal at APhO.

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 🔗 polya2005.github.io/portfolio/#

Education

BS **Massachusetts Institute of Technology**, Electrical Engineering with Computing Cambridge, MA
 • **Relevant coursework:** Computer Architecture, Digital System Lab (FPGA), Low-Level Programming, Circuits, Signal Processing, PCB Design Sept 2024 – present

Skills

Programming: C/C++, Python, RISC-V Assembly, C#, Javascript
Hardware & Embedded: FPGA, Verilog, Bluespec, Vivado, Arduino, ESP-IDF, PlatformIO
CAD & Fabrication: KiCad, Fusion 360, 3D Printing (FDM/SLA), Laser Cutting
Software & Development Tools: Git, Linux, GNU Make
Other Technical Skills: LaTeX, Unity

Experience

MIT EECS, Circuits (6.200) Lab Assistant Feb 2026 – present
 • Confirmed the students' understanding of key concepts in the labs of the Circuits course 6 months
 • Clarified class concepts to students during Lab Hours

MIT CSAIL, Undergraduate Researcher (UROP) Sept 2025 – May 2026
 • Reduced the latency of the Puppetmaster transaction scheduler by more than 90% 9 months
 • Improved the flexibility of the Bloom filter in the Puppetmaster transaction scheduler with Fibonacci hashing

MIT Game Developers Guild, Programmer Nov 2024 – Oct 2025
 • Implemented games in the Unity game engine 1 year
 • Operated in a team of writers, artists, musicians, programmers, and game designers
 • Collaborated both in person and online via Discord and GitHub

Projects

FPGA Spinthesizer Oct 2025 – Dec 2025
 • Developed an RTL design that drives stepper motors to generate different pitches at different volumes according to the MIDI input
 • Tested Verilog modules using Python and Cocotb
 • Collaborated with Git in a team of two

Star Pointing and Identification Device Mar 2021 – present
 • Developed a device that automatically points its pointer toward the star selected by the user from more than 1000 stars in the built-in database and identifies the star that the user manually points the pointer toward
 • Designed a PCB around the Raspberry Pi Pico, interfacing with peripherals via SPI, I2C, and UART interfaces
 • Modeled the enclosure using Fusion 360 and fabricated it with 3D printing
 • [GitHub repository for the first prototype](#) and [for the new, in progress version](#)